

Engineering Report

File: 2a7528879e154e79b3234bdb4747f0b8_testar-process.STEP
Assembly ID: asm_2a7528879e154e79b3234bdb4747f0_289ecc39
Generated: 2026-02-23T14:17:11

Total Parts	122
Unique Parts	67
Complexity Score	70/100
Health Score	75/100

Overview

Metric	Value
Total Parts	122
Unique Parts	67
Repeated Parts	55
Complexity Score	70/100

Bill of Materials

Item	Part Name	Qty	Material	Volume (mm³)
4	Lame1	4	N/A	400280.8
5	Pi■200	1	N/A	624622.5
6	levier100b_Seconde occurrence	1	N/A	560.6
36	ROTOR-011	1	None	2405908.8
40	But■	4	None	403621.2
41	Pi■201	1	N/A	280684.0
45	But■	4	N/A	184533.3
75	levier211	1	N/A	67117.0
106	LAME1	2	None	458457.7
107	p2-act_RNZ009	1	N/A	95798.8
110	levier100a	3	N/A	5619.3
113	levier100b001	3	N/A	527.8
114	levier100b_Seconde occurrence001	1	N/A	560.6
115	ROTOR-201NAL-Z7	1	None	21374159.9
116	BIELLE101A_type002	1	None	88066.1
120	Lame11	4	N/A	58176.7
121	SPHERE2	1	N/A	1664168.5
122	MANCHE-201	1	None	1534002.4
123	levier212	1	N/A	67117.0
124	Arbre1	1	N/A	3019956.6
125	BIELLE101A	1	None	106379.0
128	p1-act_RNZ7	3	N/A	509752.5
158	ROTOR-10	1	None	2405908.8
159	MANCHE-203	1	None	1534002.4
160	Noyeau004	1	None	4274.6

Item	Part Name	Qty	Material	Volume (mm³)
193	Butt0	4	N/A	203543.7
194	MANCHE-202	1	None	1534002.4
198	Levier1	4	N/A	85062.4
199	Noyeau002	1	None	11948.0
200	actionneur100_RNZ7	1	None	595349.0
201	Noyeau1	1	None	11948.0
203	rotule-B101A	2	N/A	16668.7
204	p1-PLATEAU_201	1	N/A	645686.6
207	levier100b	3	N/A	527.8
209	Moyeux100	2	N/A	1408851.5
212	levier100	3	N/A	5273.2
213	levier201	1	N/A	60220.5
214	PALIER-1	1	None	941153.4
215	Pi1	1	N/A	691092.2
216	Pi202	1	N/A	599741.1
218	rotule-B101A004	2	N/A	16668.7
220	levier100b_Première occurrence	2	N/A	320.3
221	p2-act_RNZ008	1	N/A	95798.8
223	LAME002	2	None	458457.7
227	Butt2	4	N/A	10163.3
229	bielle101A	2	N/A	73041.3
232	p3-act_RNZ7	3	N/A	85596.5
233	BIELLE101A_type1	1	None	88066.1
235	bielle101A_type1	2	N/A	54728.3
236	actionneur100_RNZ008	1	None	595349.0

Item	Part Name	Qty	Material	Volume (mm³)
240	Manche100	4	N/A	430392.6
242	rotule-B101A001	2	N/A	16668.7
272	p2-act_RNZ7	1	N/A	95798.8
276	Manche101	4	N/A	430392.6
277	levier210	1	N/A	36297.1
279	levier100_Première occurrence	2	N/A	3954.2
280	BIELLE101A001	1	None	106379.0
282	rotule-B101A005	2	N/A	16668.7
283	PALIER-201	1	None	771120.3
287	But1	4	N/A	189914.2
288	Noyeau2	1	None	4274.6
289	Pi203	1	N/A	161703.9
290	Noyeau003	1	None	11948.0
292	butsphere002	2	N/A	40070.4
293	MANCHE-204	1	None	1534002.4
295	butsphere1	2	N/A	40070.4
296	actionneur100_RNZ009	1	None	595349.0

Insights

Moderate Assembly Complexity (INFO)
Assembly complexity score is 70/100. Monitor for potential simplification opportunities.

Repetition Opportunity (INFO)
Part 'Lame1' appears 4 times. Consider standardizing similar parts or using modular design.

Health Check

Health Score	75/100
Warnings	1 validation error(s)

Largest Parts

Part Name	Volume (mm³)	Quantity
ROTOR-201NAL-Z7	21374159.9	1
Arbre1	3019956.6	1
ROTOR-011	2405908.8	1
ROTOR-10	2405908.8	1
SPHERE2	1664168.5	1

Repetition Analysis

Repeated parts account for 68.9% of total parts.

Part Name	Quantity
Lame1	4
But■	4
But■	4
Lame11	4
But■0	4
Levier1	4
But■2	4
Manche100	4
Manche101	4
But■1	4

Manufacturing Hints

High Part Repetition

87.2% of parts are repeated. Consider batch manufacturing or modular tooling.

Large Part Detected

Largest part 'ROTOR-201NAL-Z7' has volume 21374160 mm³. Consider manufacturing method and material selection.

Next Steps

1. Review assembly snapshot and BOM for accuracy
2. Verify part quantities match design intent
3. Evaluate manufacturing strategy for repeated parts

Reference Geometry (Excluded from Manufacturing Analysis)

The following 6 item(s) were identified as reference geometry and excluded from volume calculations, largest parts analysis, manufacturing hints, and complexity scoring.

Part Name	Reason
Y-axis	name contains reference geometry keyword
X-axis	name contains reference geometry keyword
YZ-plane	name contains reference geometry keyword
Z-axis	name contains reference geometry keyword
XZ-plane	name contains reference geometry keyword
XY-plane	name contains reference geometry keyword